

F24-185-D-Smart gallery

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Chapter 1

Introduction

Smart Gallery is a final year project dedicated to developing an intelligent image management and retrieval system that enhances user interaction with digital photo galleries using Artificial Intelligence (AI) and Natural Language Processing (NLP). In today's era of vast digital imagery, efficiently organizing and retrieving images is a significant challenge. Traditional gallery apps often lack the sophistication to interpret natural language queries or categorize images beyond basic metadata.

This project addresses these limitations by introducing advanced features such as person and category classification, geo-location tagging, natural language-based search, and image editing based on textual prompts. By integrating these AI-driven functionalities into a mobile application, **Smart Gallery** ensures accessibility and convenience for users across all professions.

Through this Software Requirements Specification (SRS) document, we aim to provide a comprehensive understanding of the objectives, functionalities, and technical aspects of **Smart Gallery**, ultimately empowering users to navigate their digital media with greater ease and efficiency.

1.1 Problem Statement

The exponential growth of digital images has made it increasingly difficult for users to manage and retrieve photos from large galleries. Traditional gallery applications offer limited functionalities, relying on basic categorization and manual tagging, which are time-consuming and inefficient. As a result, users struggle to find specific images based on complex criteria such as particular individuals, objects, scenes, or sentiments, leading to frustration and decreased productivity.

Additionally, there is a lack of intuitive search options using natural language, and exist-

ing tools do not effectively extract meaningful information like text or geo-location from images. The absence of integrated intelligent editing features and seamless mobile accessibility further hampers the user experience, leaving users unable to fully engage with their digital memories.

Smart Gallery aims to address these challenges by developing an AI-powered image management and retrieval system. By introducing advanced features like automatic classification, natural language search, information extraction, and intelligent editing within a user-friendly mobile application, **Smart Gallery** enhances user interaction and efficiency. This solution empowers users to navigate their digital media with greater ease, making the management of vast photo collections a seamless and intuitive experience.

1.2 Scope

The project's scope encompasses the development of "**Smart-gallery**" a user-friendly and versatile mobile application designed to simplify the integration of Classification , Diffusion and text to image based Models into edge devices.

1.3 Modules

The Smart Gallery application is structured into several modules, each focusing on specific functionalities to enhance image search and retrieval through natural language inputs.

Below is a detailed description of each module and its associated features.

1.3.1 Image Classification

This module utilizes state of art classification model to automatically classify each unique person within solo or group images , predicting there age and emotion within image .

1. Person Classification

- Identify individuals in images.
- Predict age.
- Detect facial expressions to assess emotion.

2. Category Classification

- Recognize objects such as cars, animals, buildings and everyday items within the background of image .

- Classify scenes like landscapes, urban settings, and indoor environments.
- Geo-location tagging , to categorize images by location at which they are taken.
- Automatically assign images to relevant categories.

1.3.2 Search and Retrieval

This module provides text based search capabilities, allowing users to find images using natural language queries or by providing a reference image.

1. Text-Based Image Search and Retrieval

- Enable searches using natural language prompts.
- Support complex queries involving multiple criteria.
- Provide quick and accurate search results.

2. Reverse Image Retrieval

- Search for similar images using an input image.
- Detect duplicates or related images within the gallery.

1.3.3 Caption Generation

This module uses state of the art LLM to generate Creative and catchy captions for images, Easing the life for social media animals .

1. Caption Generation

- Generate fancy and creative captions based on image content.
- Support caption customization and editing.

1.3.4 Image Editing

This module allows users to edit and style images using natural language prompts, offering diffusion model editing capabilities within the app.

1. Prompt-Based Inpainting

- Edit images by describing desired changes in text.

- Remove or replace objects within images.

2. Style Transfer and Blurring

- Apply artistic styles to images.
- Blur specific areas to obscure sensitive content.

1.3.5 Mobile Application Integration

This module ensures that all functionalities are seamlessly integrated into a user-friendly mobile application that is connected to user gallery , enhancing accessibility and user engagement.

1.4 Constraints

- **Resource Limitations:** As a undergrad students project, Smart Gallery operates under limited resources in terms of time, budget, and personnel. Meanwhile ,these factors can impact the project’s overall scope, depth, and execution.
- **Technical Constraints:** Implementing advanced algorithms and LLM based functionalities such as image recognition, natural language based image searching editing require significant technical expertise.
- **Computational Resources:** LLM powered features require substantial computational power for training models and real-time processing. Limited access to high-performance computing infrastructure may affect the performance, speed, and scalability of the application.
- **Integration Challenges:** Integrating various AI components into a mobile application poses technical and hardware difficulties, especially when ensuring compatibility and performance across different edge devices .

1.5 Out of Scope

The following are not included in the scope of this project:

- Support for iOS or other non-Android platforms.
- Integration with social media platforms beyond basic sharing functionalities.
- Data privacy features or compliance with specific data protection regulations.
- Custom AI model training on user-provided datasets

1.6 User Classes and Characteristics

The Smart-gallery system is designed to cater to a variety of user classes, each with distinct roles and responsibilities. Below are the anticipated user classes and their pertinent characteristics.

User class	Description
Casual Users	Individuals who use their smartphones primarily for personal purposes, capturing everyday moments, events, friends, and family. They may not be technical experts but appreciate features within their gallery that simplify tasks. These users benefit from categorized photo albums, easy retrieval using natural language queries like "photos of me at the beach."
Professional Photographers	Users who take high-quality images for professional purposes, such as photographers, graphic designers, and artists. They often manage large volumes of images and require advanced organizational and editing tools. They require categorizing people and backgrounds, reverse image search to find similar or duplicate images, and prompt-based image editing to make quick adjustments.
Social Media Influencers	Individuals who frequently post on social media platforms and seek creative captions and editing tools to enhance their online presence. They require quick caption generation to create engaging posts, style transfer or image editing tools on the go to enhance content posting.
Travelers	Individuals who travel frequently and accumulate a large number of photos from different locations from different trips. They enjoy documenting their experiences and sharing them with others. This class of users benefit through geo-location tagging, categorization of landmarks, and generating captions that include location-specific information. Moreover, they can use natural language search to revisit memories by querying "sunset photos in Bali."

Chapter 2

Project Requirements

2.1 Use-case

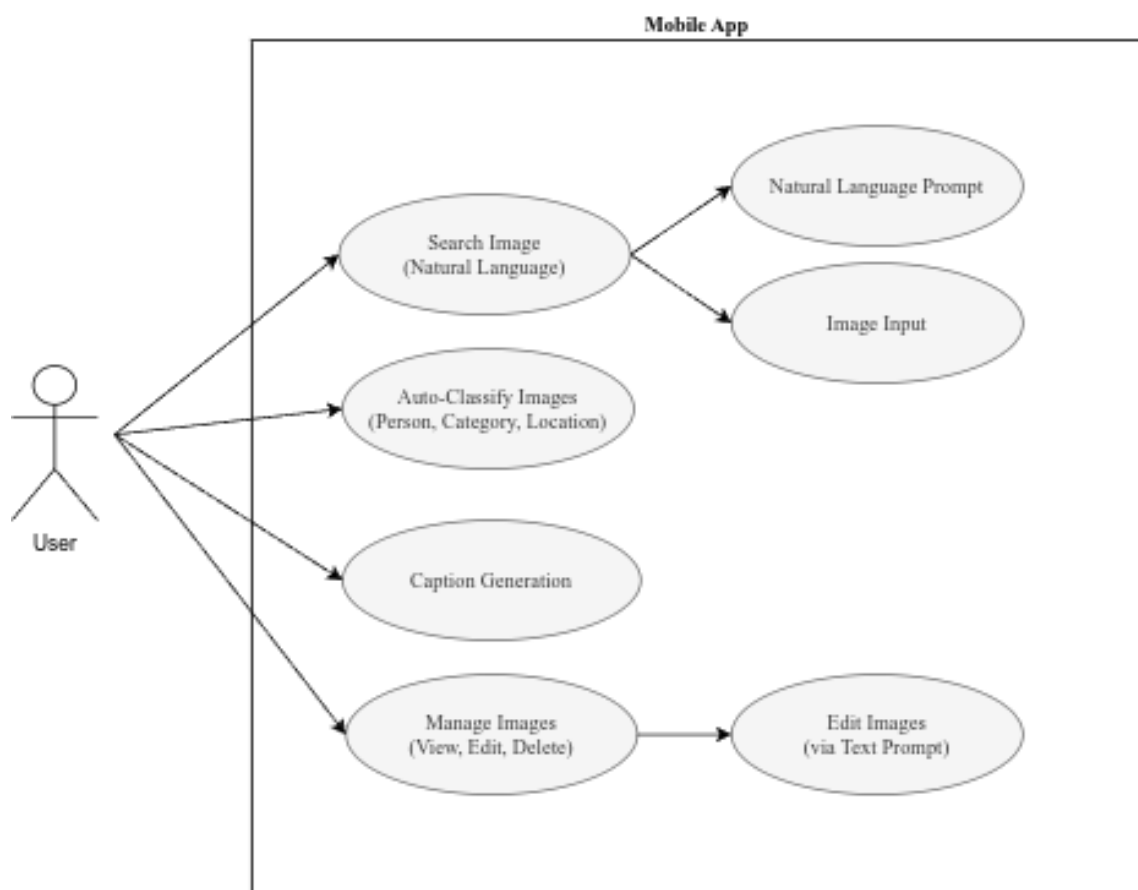


Figure 2.1: Use-Case Diagram

2.2 Functional Requirements

The "Smart Gallery" project incorporates a range of functional requirements to deliver an intelligent and user-friendly mobile application that enhances image management and retrieval through AI-powered features. These requirements ensure the successful development and integration of various modules, including image classification, search and retrieval, caption generation, and image editing. The core functional requirements are as follows:

2.2.1 Module 1

Following are the requirements for module 1:

1. **Person Classification:** Automatically detect and identify individuals in images, predicting attributes like age and facial expressions.
2. **Category Classification:** Recognize objects such as vehicles, animals, or buildings, and classify them into relevant categories.
3. **Scene Recognition:** Detect and classify environments such as landscapes, urban settings, or indoor scenes.
4. **Automatic Captioning:** Generate descriptive, creative, and user customizable captions based on the content of the image.
5. **View, Edit, and Delete Images:** Enable users to view images in their gallery, apply edits, and delete images as needed.

2.2.2 Module 2

Following are the requirements for module 2:

1. **Text-Based Search:** Enable users to search for images using natural language prompts, supporting complex queries.
2. **Reverse Image Search:** Allow users to upload an image to find visually similar or related images in the gallery.
3. **Prompt-Based Inpainting:** Allow users to modify images by describing the desired changes in text, such as object removal or replacement.
4. **Mobile Application Integration :** Develop a simple, intuitive, and responsive UI for easy navigation and interaction across all features.

5. **Natural Language Interaction:** Facilitate smooth, natural language interactions for image searches, editing, and management.

2.3 Non-Functional Requirements

2.3.1 Performance

- The system must load and process images quickly, ensuring smooth performance even for large galleries.
- AI-powered features like image classification and caption generation should deliver results in a timely manner for each image.
- The app should handle a substantial number of images without noticeable degradation in performance.

2.3.2 Usability

- The user interface must be intuitive and easy to navigate, requiring minimal user training.
- All key features such as search, edit, and classification should be accessible within 2 clicks or taps.
- The mobile app should be designed for both beginner and advanced users, with simple tutorials available

2.3.3 Scalability

- The system must be scalable to support additional features, such as video classification, without requiring a full redesign.
- The database and AI model infrastructure must support the addition of new categories, locations, and classification criteria with minimal performance impact.

2.3.4 Compatibility

- The app must work on all modern Android devices and must be adaptable for future versions.

- The system must be compatible with standard image formats, including JPEG, PNG

2.3.5 Reliability

- The system should maintain a high level of availability to ensure continuous access for users.
- All operations, including image uploads, classification, and edits, must be accurately completed without data loss.
- A backup mechanism must ensure that no user data is lost, even in the case of server failure.

2.3.6 Privacy

- Users' images and interactions must be kept private, with clear terms of use and privacy policies.
- The app should allow users to control the level of sharing, ensuring full control over their data.

Chapter 3

System Overview

Smart-gallery is designed to cater to multiple features that required different image model processing that are leveraged and combined to develop this 1 in all gallery app. The architectural and design Models shown below illustrate the working of each feature.

3.1 Architectural Design

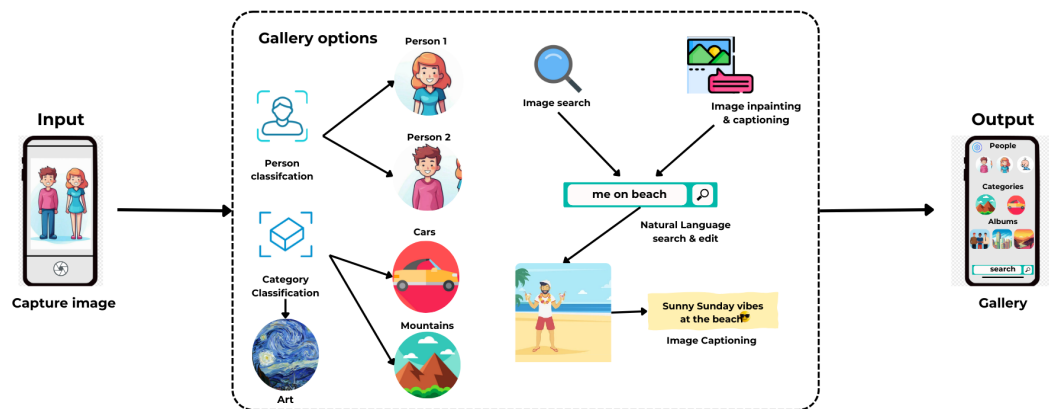


Figure 3.1: Architecture

3.2 Design Models

3.2.1 Activity Diagrams

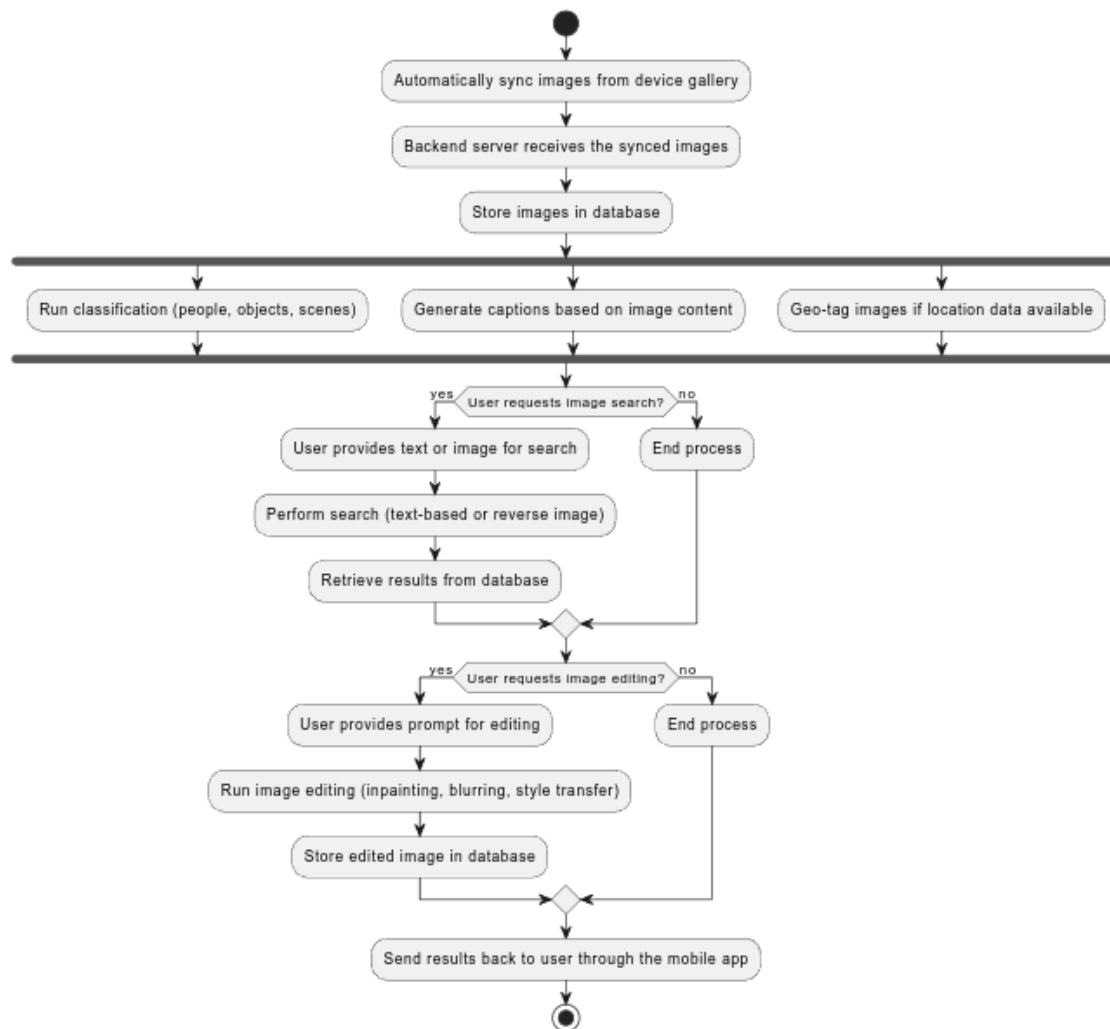


Figure 3.2: System Activity Diagram

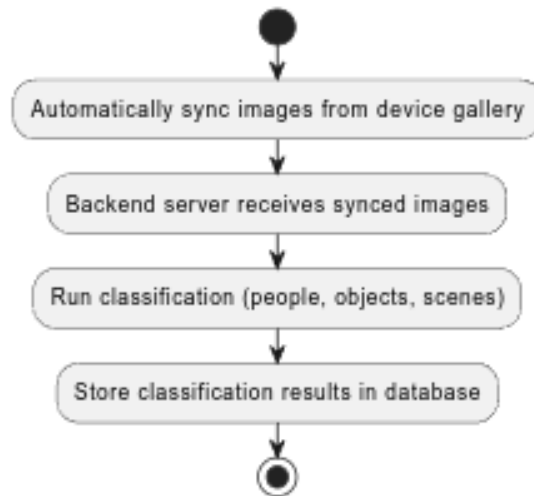


Figure 3.3: Classification Activity Diagram

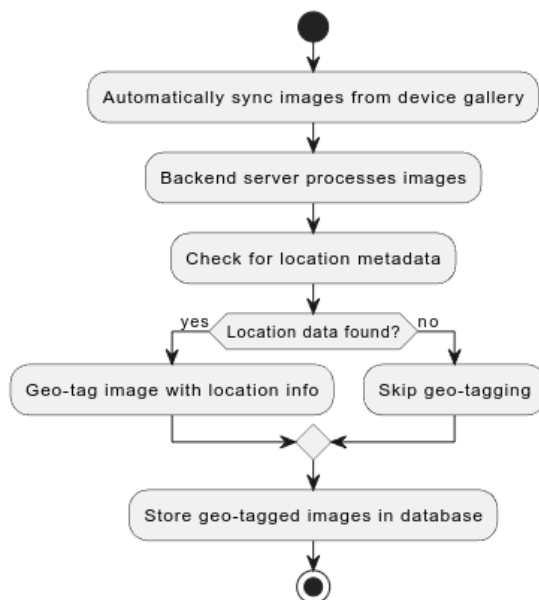


Figure 3.4: Geo-Location Tagging

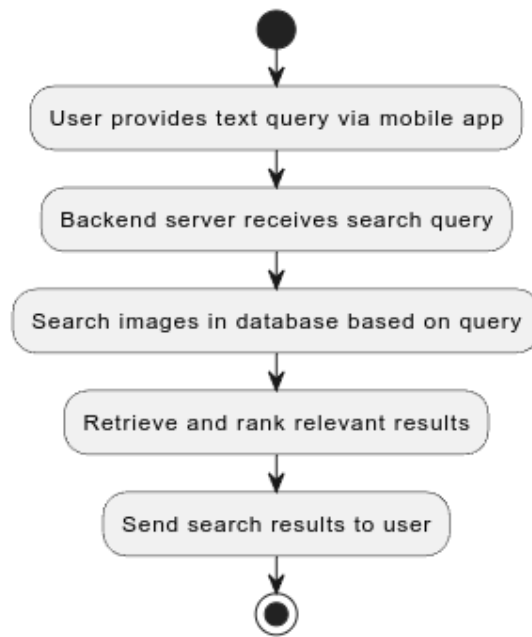


Figure 3.5: Text Query Searching

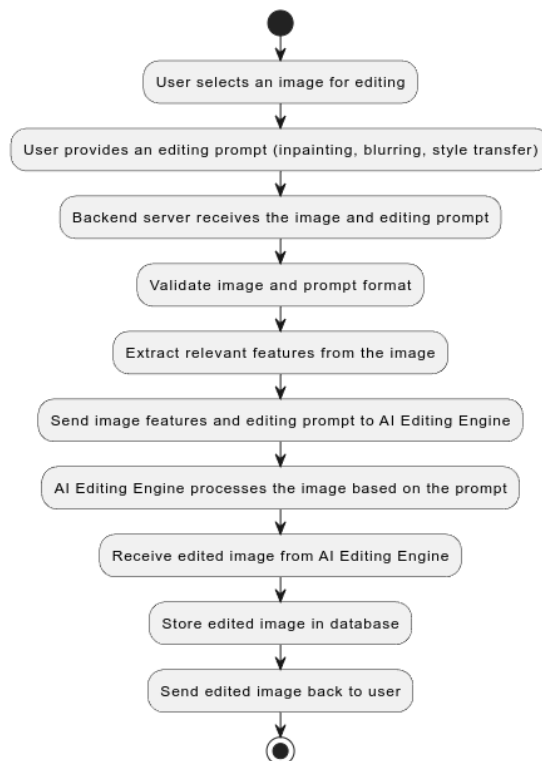


Figure 3.6: Prompt Editing Activity Diagram

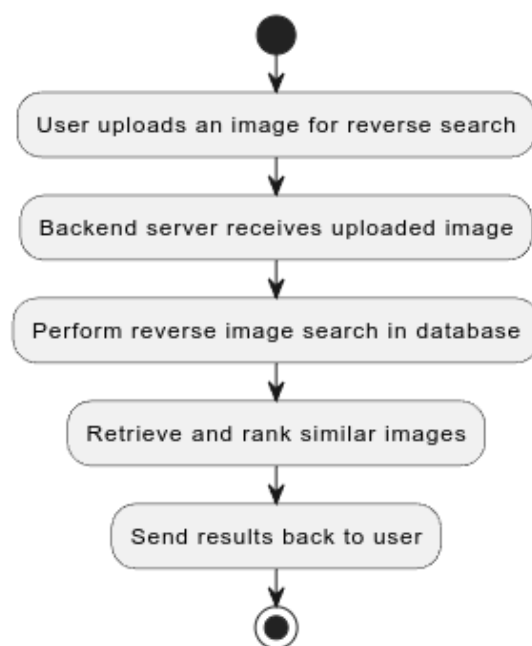


Figure 3.7: Reverse Image Search

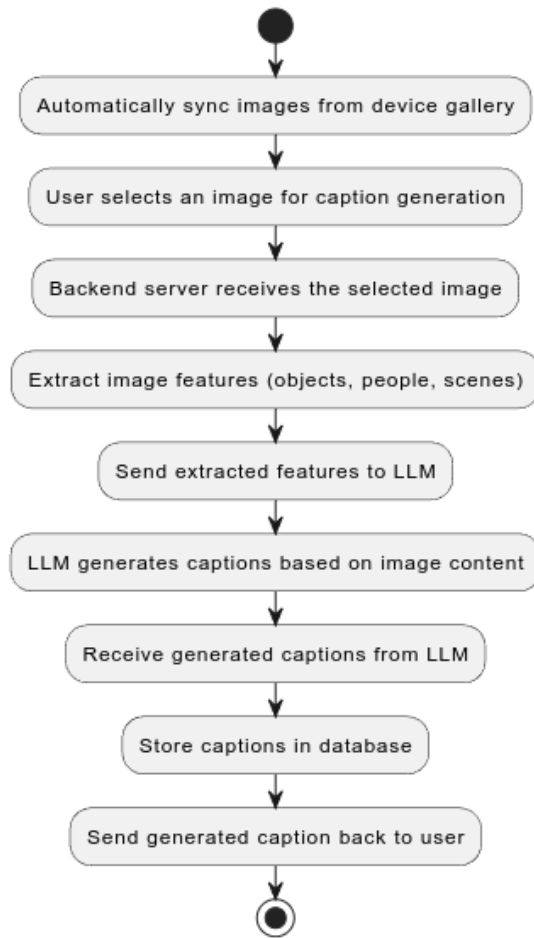


Figure 3.8: Caption Generation Activity

3.2.2 Class Diagram

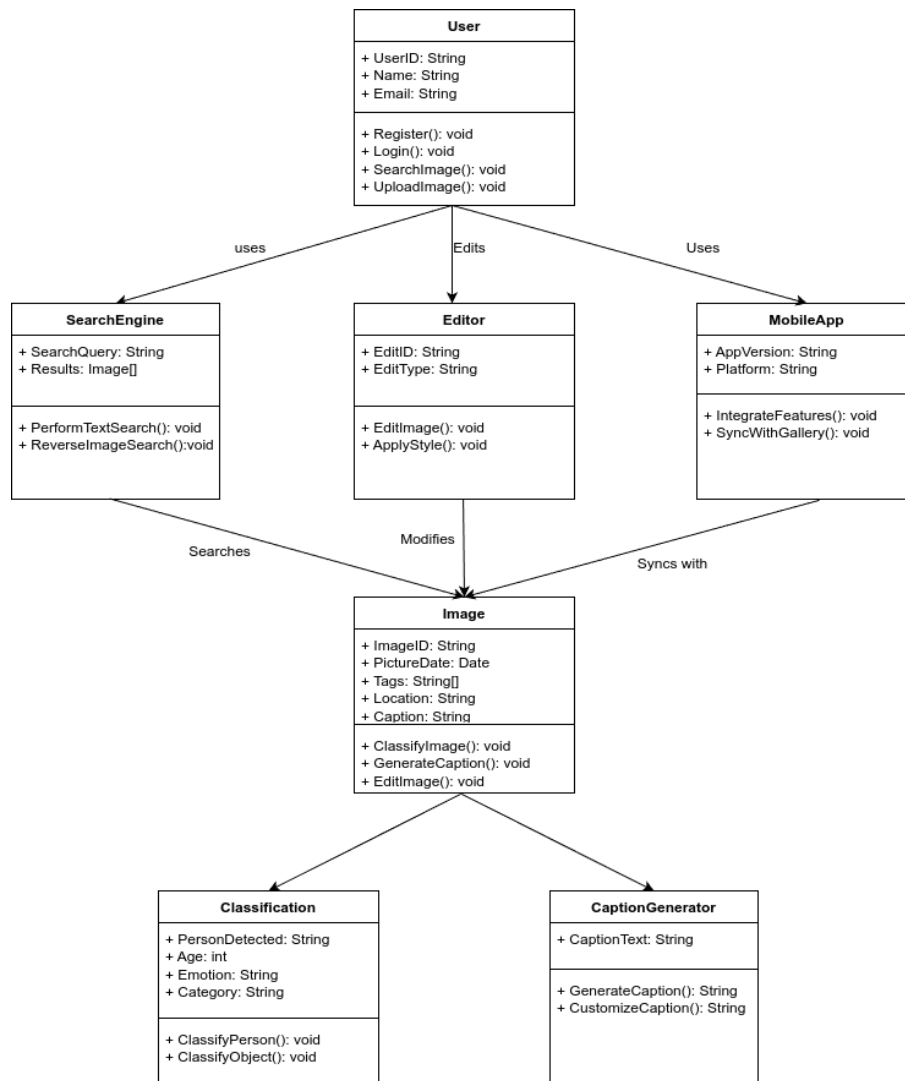


Figure 3.9: Class Diagram

3.2.3 Class-level Sequence Diagram

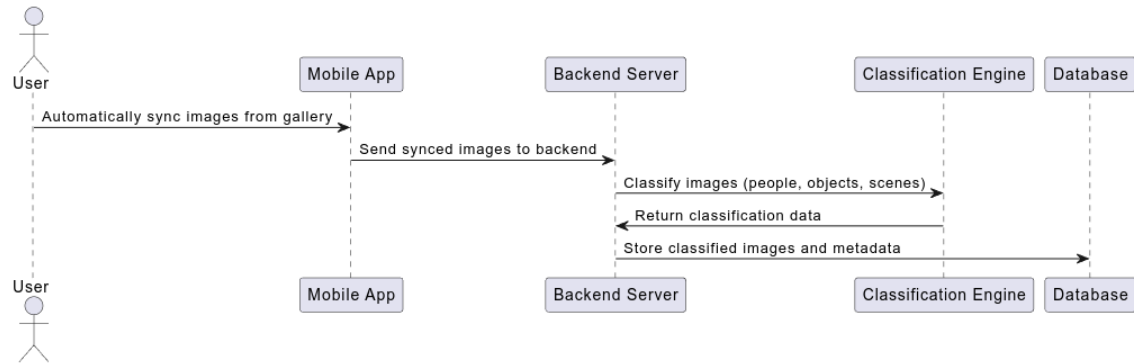


Figure 3.10: Person and Object Classification

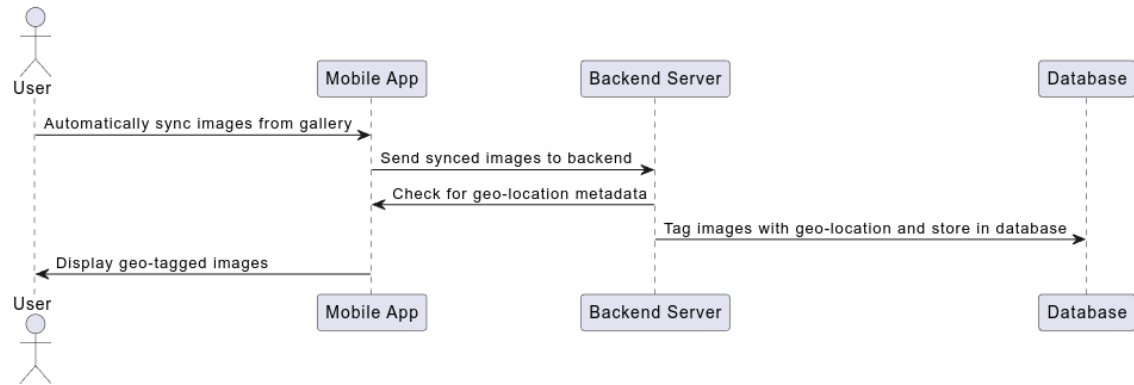


Figure 3.11: Geo-location Tagging

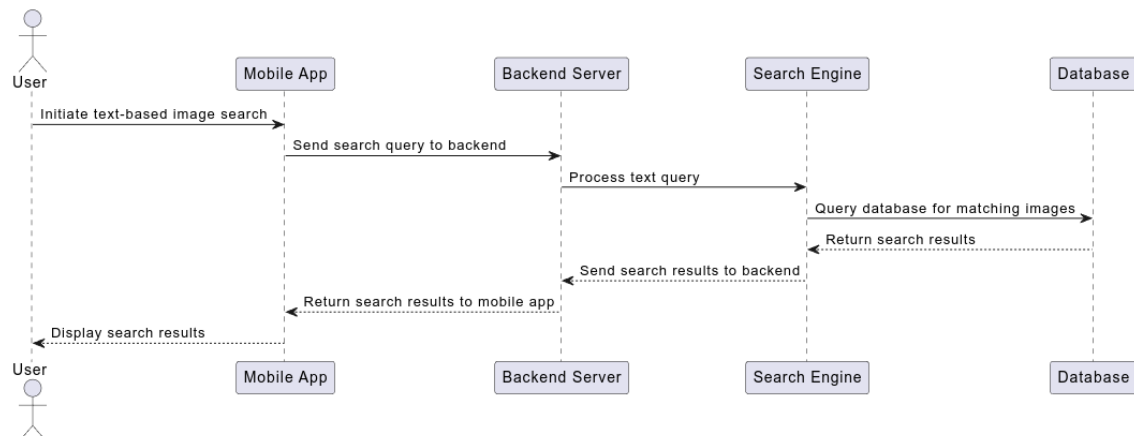


Figure 3.12: Text Query Searching

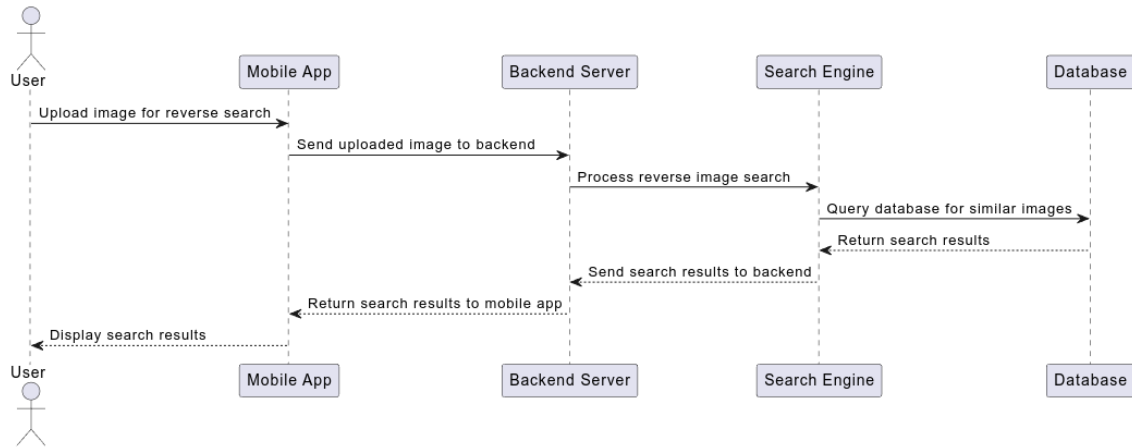


Figure 3.13: Reverse Image Retrieval

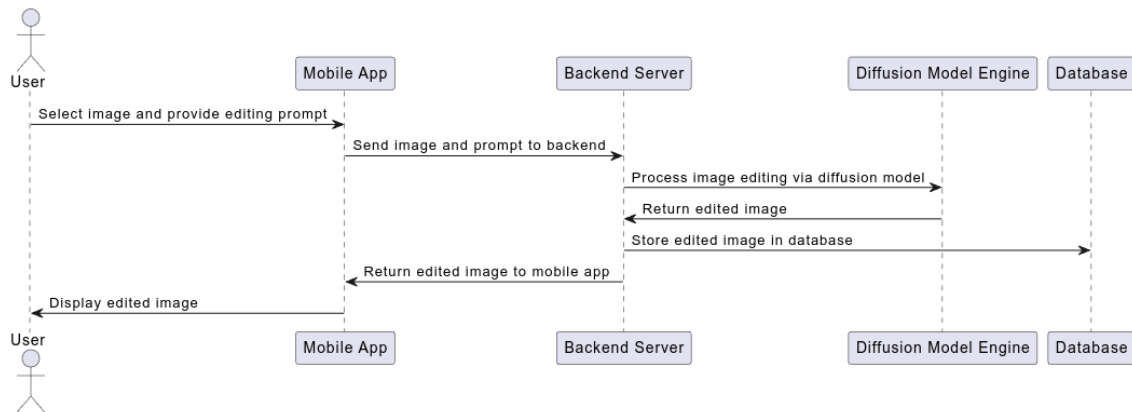


Figure 3.14: Prompt-based In-painting

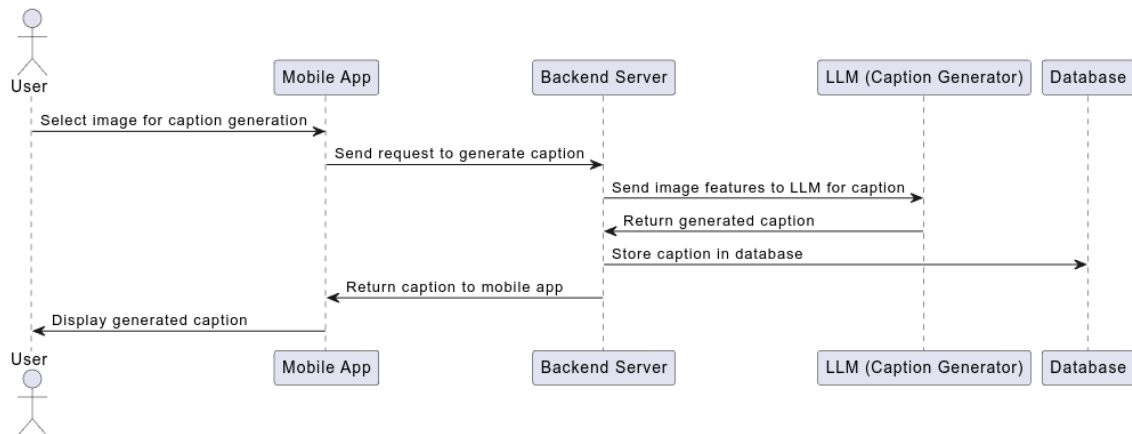


Figure 3.15: Caption Generation

3.3 Data Design

This section explains how the information domain of the Smart-gallery system is transformed into data structures. It describes how the image data is stored, processed, and organized. Additionally, it lists the databases or data storage items used in the system.

3.3.1 ER-Diagram Explanation

The ER diagram illustrates the data model for the Smart-gallery system, highlighting key entities and their relationships. Below is a detailed description of each entity and its attributes:

- **USER** Attributes: id, name, email Relationships: has (1:1) with Search-Engine, has (1:1) with Editing, has (1:1) with Mobile-APP
- **Search-Engine** Attributes: Search-Query, Result-Image Relationships: has (1:1) with Images,
- **Editor** Attributes: Edit Query, Edit-type Relationships: has (1:1) with Images
- **MobileApp** Attributes: App-version, platform Relationships: has (1:1) with Images
- **Images** Attributes: id, Date, Tags, Location, Caption Relationships: has (1:1) with image classification , has (1:1) with Caption generator.

3.3.2 Database

The Smart-Gallery system uses SQLite and Milvus as its database management system. SQLite is a lightweight, file-based database that provides efficient storage and retrieval of data. Milvus is also a lightweight, vector database that provides efficient storage and retrieval of vector data .

Both databases are suitable for applications where simplicity and minimal setup are required.

The key features of SQLite include:

- Self-contained: No external dependencies or server setup is needed.
- Serverless: Operates directly on the file system without requiring a separate server process

- **Zero Configuration:** No configuration is necessary, making it easy to integrate and use.
- **Cross-platform:** Works consistently across various operating systems.
- **Efficient:** Optimized for read-heavy operations with low memory overhead.

The key features of Milvus include:

- **Vector Database:** it is specifically designed to handle vector data, which is used in AI applications like image, video and natural language processing.
- **Scalability:** It supports horizontal scalability, allowing to handle large-scale data efficiently .
- **High Performance:** Optimized for fast retrieval of vectors, it offers low-latency search even with billions of vector data points.
- **Open Source:** Milvus is open-source, with a large community and frequent updates, making it a reliable option for long-term projects.

The combination of SQLite and Milvus is used to store image data and the embeddings generated by the processing models used within our application such as text based searching and editing ,Caption generation and image classification tags.

which are crucial for maintaining consistent states and effiecient processing within the Smart-gallery system.

Bibliography